hooks folder stores all custom hooks used in the project. This flat structure is effective for small, focused projects. However, it may not scale well for larger or more complex applications. I'll provide alternative project structures with better scalability when building more intricate features in future chapters.

If you experience problems with the AI not responding to your requests when using OpenAI, it is important to ensure that your OpenAI account has sufficient credits before making API calls; otherwise, you will encounter an `insufficient_quota` error. Additionally, you cannot use the same phone number with different email addresses to obtain additional free credits on OpenAI. If you have already used a free trial, the credits are shared across accounts linked to your phone number. This limitation applies specifically to OpenAI accounts and does not affect the use of other providers such as Google Gemini. Following these guidelines will help you avoid common quota-related errors and ensure smooth API usage with OpenAI.

2.3 Under the hood: The generative AI lifecycle

Let's explore the mechanics that occur when you interact with Astra, our conversational AI agent. Behind the scenes, as you can see in figure 2.4, your message travels from the chat interface to the backend, where it interacts with the AI model and triggers a response. The backend then delivers that response, which updates the chat window and allows you to continue the conversation with Astra. While doing so, the UI displays a loading indicator for the user. Once the new message arrives, the page automatically scrolls to display the last response message.

Some interactive elements are working behind the scenes without user intervention, so we'll want to start with the rules we follow.

2.4 Designing for a better user experience

As we explore the interactivity of our generative AI application, it's important to consider the user experience (UX) when dealing with long responses from LLMs. Here's why:

- *Attention management*—Lengthy, unformatted text blocks can overwhelm users and hinder information processing. To address this, many LLMs allow you to specify both the desired response format through prompts and their delivery method using streaming responses. Requesting content in formats such as Markdown allows for easy rendering with improved readability and structure. This enables features like collapsible sections, code blocks, and bolded text within the application instead of showing a big wall of text.
- *Navigation and control*—Users might struggle to navigate within lengthy responses, so consider adopting automatic scrolling mechanisms or ways to scroll to the top or bottom of the message threads.
- *Engagement and interactivity*—Providing users with ways to interact with the response, like collapsing sections or highlighting keywords, can enhance engagement and understanding.